Rhetoric and Resolution: Using UN Speech to Predict Voting Behavior on Palestine

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Abstract. This study investigates whether UN General Debate speeches can predict voting on Palestine-related resolutions, supporting the 80th UN debate theme and SDG 16. Using the UN General Debate Corpus, voting records, sentiment scores, frequency of mentions, and contextual factors (region, recognition status), analysis found geography and mentions of Palestine predicted support more than sentiment. Models confirmed this but struggled with class imbalance, highlighting the importance of structural alliances over political rhetoric.

Keywords: Palestine · UNGD · Voting behavior · Sentiment analysis · Random Forest

1 Introduction

Every year the United Nations hosts the General debate with an overall theme, in 2025 the theme is "Better together: 80 years and more for peace, development and human rights". According to United Nations, 2025a, the General Debate provides an opportunity for member states to articulate their positions on global issues through annual speeches.

In 2015 the United Nations (UN) collectively adopted 17 different Sustainable Development Goals (SDGs) to promote peace and prosperity for the world (United Nations General Assembly, 2015). These goals play a prominent role in the UN General debate (United Nations, 2025b). Goal 16 is promoting peaceful and inclusive societies, providing access to justice for all and building effective, accountable and inclusive institutions at all levels (United Nations General Assembly, 2015). The humanitarian crisis in Gaza represents a critical global issue impacting this goal and the broader debate theme of peace and human rights, with Bouri (Bouri, 2024) noting its substantial impact on both the region and the international community.

The "Question of Palestine" has been regularly discussed and managed by UN throughout almost all of its 80 year existence (United Nations, n.d.). This has led to large amounts of resolutions regarding Palestine, which are formal, legally binding directives adopted by the UN to guide member states' actions and influence international policy.

Despite the extensive history of UN resolutions concerning Palestine, accurately predicting members votes on new resolutions remains challenging. Approaches often rely on historical voting patterns, alliances, or geopolitical analysis like those proposed by Zaghi, 2021, which may not fully capture shifts in national attitudes that may been given by official speeches. These speeches provide a valuable language resource that provide insight into the diplomatic stance of nations and the sentiment on critical issues such as the humanitarian crisis in Gaza. Understanding these relationships may be useful for policymakers and international organizations seeking to anticipate voting behavior and design effective diplomatic strategies.

This study combines exploratory data analysis (EDA) and Natural Language Processing (NLP) techniques to analyse the attitude of UN General Debate speeches. By evaluating the positive or negative language in these speeches as sentiment scores and measuring the frequency of Palestine-related references, it becomes possible to find patterns that may predict voting outcomes for resolutions on Palestine.

Through the combination of these speeches with historical voting data, region, year, and whether the country had officially recognized Palestine, the study aims to develop a predictive model that improves understanding of global attitudes toward Palestine and predicts how UN members are likely to vote in the future.

The research asks two specific questions: Exploratory question: Are there patterns of Palestine mentions present in UN General Debate speeches, geography and state recognition and how do these relate to the voting behavior of nations? Hypothesis: Countries with close ties to Palestine through, geography, diplomatic sentiment, and recognition are more likely to vote favorably in the UN towards Palestine, Predictive question: Can a country's speech along with information such as its region and state recognition accurately predict voting behavior on Palestine-related resolutions? Hypothesis: Using speech sentiment scores and contextual features in a predictive model will enable accurate prediction of voting patterns.

This report is structured as follows: Section 2 details data preparation and exploratory analysis, Section 3 presents results and discussion, and Section 4 concludes the study.

2 Methodology

2.1 Data Sources This study combines multiple datasets to investigate the relationship between diplomatic discourse and voting behavior in the United Nations (UN) on Palestine-related resolutions.

1. **UN General Debate Corpus (UNGDC) (Jankin et al., 2017):** Annual statements delivered by UN member states between 1946 and 2024. The speeches provide a consistent record of states' diplomatic stances and represent a key qualitative input into the analysis.

2. **UN General Assembly Voting Records (United Nations Dag Hammarskjöld Library, 2025):** Official resolution votes from the UN General Assembly that consists the votes of every UN member state for each resolution from 1946 till 2025.
3. **Palestine Recognition Dataset (World Population Review, 2025):** A supplementary dataset recording the years in which states formally recognized Palestine was incorporated to provide further context for both speech sentiment and voting behaviour.
4. **Standard Country or Area Codes (United Nations Statistics Division, n.d.):** A dataset with the area codes of countries to provide geographic context to our data.

Together, these sources capture both rhetoric and actions of the UN towards Palestine, enabling the creation of a cohesive dataset merged using the ISO alpha-3 codes of nations and session numbers to perform analysis to investigate the hypothesis of the study.

2.2 Data Preparation A structured pipeline was implemented to prepare the data for analysis:

Text Cleaning and Tokenization: The UN General Debate speeches often contain punctuations, capitilisation inconsistencies and unusual formatting. To standarize the text, punctuations were normalized, text was converted to lowercase non-alphanumeric characters were removed, and speeches were tokenized into sentences, thus reducing any unnecessary noise. This facilitated both keyword filtering and sentiment scoring.

Filtering of Speeches: A keyword approach was used to find references to Palestine. This offers a transparent, reproducible way to isolate Palestine-related discourse. The keyword dictionary was made iteratively and included the terms:*Palestine*, *Gaza*, *West Bank*, *PLO*, *East Jerusalem*, *two-state*.

Text normalisation and TF-IDF term selection: A stopword list combined NLTK's English stopwords with subject stopwords (e.g., united, nations, assembly) and the Palestine keyword stems to avoid these frequent but neutral tokens being the most frequent terms. TF-IDF (Bird et al., 2009) was computed across all Palestine sentences and the top 200 terms by mean TF-IDF score were stored as frequently used terms. The TF-IDF step served two purposes. Firstly to understand frequently used vocabulary for reporting, secondly to identify domain terms to extend VADER sentiment lexicon.

Sentiment lexicon augmentation and scoring: A sentence sentiment score was made using VADER Sentiment Intensity Analyzer (Hutto & Gilbert, 2014). Since VADER's lexicon does not cover diplomatic or region specific terms, the TF-IDF terms absent from VADER were matched to SenticNet polarity intensities, following a similar approach to Bau et al., 2023 that was applied in sentiment analysis of customer reviews. SenticNet polarity (range -1 to 1) was rescaled to the VADER numeric range by multiplying by 4 (resulting in a -4 to 4 scale) and appended to the VADER lexicon. A small exclusion list (e.g., world, general, countries) prevented non-informative additions. In total,

72 TF-IDF terms were incorporated into the VADER lexicon. Compound scores were computed for each Palestine sentence and aggregated to a speech sentiment metric by arithmetic mean. Speeches with no Palestine mentions were coded as neutral. For classification of polarity, VADER's standard thresholds were used: compound $x > +0.05$ positive, compound $x < -0.05$ negative, otherwise neutral.

Voting dataset filtering: The voting dataset was filtered to keep Palestine related resolutions using the Palestine keywords mentioned in 2.2 in Text Cleaning and Tokenization. Furthermore, until 2025, Palestine cannot vote in resolutions. For this reason, only resolutions where Israel voted "No" were kept, to ensure that countries voting "Yes" in these resolutions would support Palestine.

Merging Datasets: For each country and year observation, speech sentiment, frequency of Palestine references, voting records, and recognition status were combined into a single dataset. All datasets were standardized to ISO alpha-3 country codes. After joining, missing values were low in number and were mostly due to nations that no longer exist which were removed. From the period of 1946 to 1968 there were very few resolutions regarding Palestine, so they were excluded. This pipeline ensured that the analytical dataset was both reproducible and comprehensive.

2.3 Exploratory Data Analysis Exploratory data analysis was performed to try to find patterns between UN speeches and their relation to voting outcomes. First, a comparison between average sentiment and Palestine references was conducted throughout the years to examine the correlation between rhetorical tone and salience over time. Moreover, another visualization was made on vote behavior per country to show the variance of Palestine support throughout regions. Lastly, to find the correlation between voting outcomes between Palestine mentions and states that have recognize Palestine, two time series were created respectively. These exploratory steps guided feature selection for the predictive modeling stage.

2.4 Feature Engineering To effectively link sentiment, Palestine mentions, Region, Palestine recognition, and Year with voting outcomes, additional preprocessing to variables was implemented. First, taking into account that multiple resolutions take place each Session, the assumption is made that a country is in favor of Palestine for a given year if they voted "Yes" in more than 50% of Palestine related resolutions. Thus, "Yes" is represented with 1, while "No/Other" with 0. Additionally, one-hot encoding is implemented for categorical variables of Region name, Palestine recognition, sentiment label, as well as for the year to treat the variable as a distinct geopolitical context. Furthermore, using the boxplot (Fig 4) that illustrates the distribution of mentions of Palestine for supporting and non supporting countries, clear outliers can be detected in both groups (USA and Israel for non supporting, Jordan and Egypt for supporting countries) and exclude them from model training.

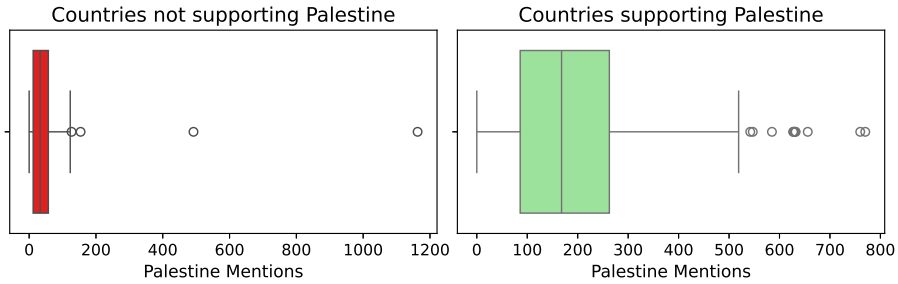


Fig. 1: Boxplots of Palestine Mentions by Voting Behavior

2.5 Predictive Modeling The dataset was divided into 80% for training and 20% for testing. To achieve this, the votes up to 2015 were used for training, and the votes for the subsequent 8 years were used for testing. For this classification problem, the performance of Gaussian Naive Bayes, Logistic Regression, and Random Forest Classifier was evaluated, setting the parameter of "class_weight" to "balanced" to pay more attention to the imbalanced class. Due to this imbalance, more importance was given in the F1-score for the imbalance class, rather than accuracy. Based on the results of the table, Random Forest outperforms the rest of the models in the F1-score of label 0.

Table 1: Baseline models with accuracy and F1-scores for both outcomes

Model	Accuracy	F1-score (Yes=1)	F1-score (No=0)
Random Forest	0.837	0.90	0.53
Naive Bayes	0.844	0.91	0.44
Logistic Regression	0.518	0.59	0.42

Subsequently, Grid Search is applied to identify the optimal hyper-parameters for the Random Forest model, increasing the f1-score for minority class to 54%.

3 Results and Discussion

In order to gain an understanding of how the data relates to the purposed hypotheses introduced in section 1 EDA needed to be conducted. The EDA was to describe time, spatial, and political patterns.

Figure 2 shows that mentions of Palestine peaked around the late 1970s–early 1980s, aligning with major conflicts and diplomatic milestones, and shows a major spike in 2023 aligning exactly with the invasion of Gaza in 2023 (The Editors of Encyclopaedia Britannica, 2025). This shows that key events in Palestine have a direct impact on the mentions of Palestine in the UN, which could lead to a

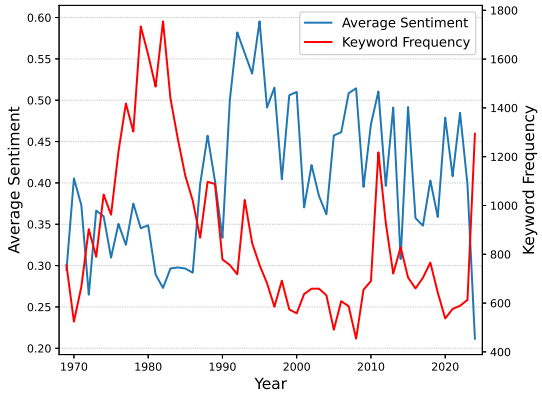


Fig. 2: Average sentiment and mentions of Palestine in UN (1969-2024)

direct impact on voting. The sentiment of the speeches is also directly inversely correlated to the mentions. This could be due to either use of language that is negative surrounding Palestine such as the use of 'explosives' within the region or the use of negative words in support such as the acknowledgment of the 'exploitation' of the Palestinian people. Highest scoring TF-IDF words from speeches on Palestine shown in appendix A 4.

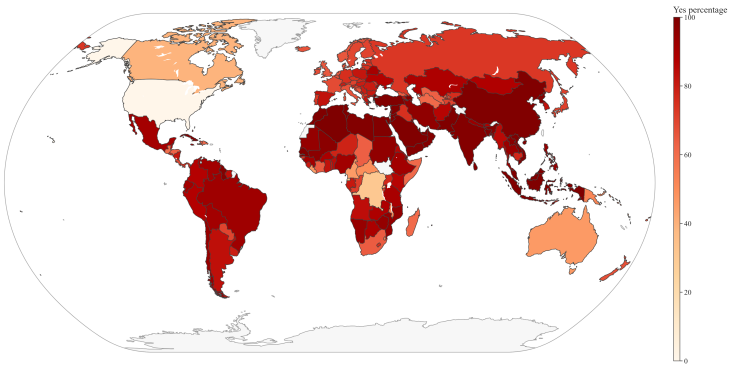
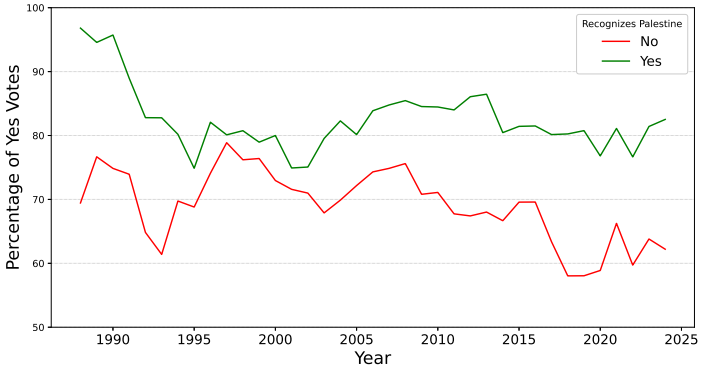


Fig. 3: Percentage of Voting for Palestine per country (1969-2024)

Figure 3 displays that the support for Palestine-related UN resolutions is highest across Africa, Asia and Latin America. While it appears to be less in North America, Europe and Oceania. This shows that geography plays a critical

role in voting on Palestine related resolutions with those closest to Palestine voting most favorably, Israel being the only exception. Shown in 4.

Figure 4 contains two crucial plots which demonstrate the clear relationship between recognition of Palestine and mentions within the UN and voting yes in the UN. It can clearly been seen those who don't recognize Palestine and those who do not mention Palestine are far less likely to vote in support of Palestine.



(a) Percentage of Yes votes for Palestine

(b) Mentions of Palestine

Fig. 4: Comparison of recognition and mentions vs. voting models

Therefore it can be concluded that UN members with close ties to Palestine through, geography, diplomatic sentiment, and recognition are more likely to vote favorably in the UN towards Palestine, confirming our exploratory hypothesis.

For the predictive hypothesis of the research, the aim of was to develop a model that will be able to predict whether a country voted in favor or against Palestine on a specific year, based on its region, the number of mentions of Palestine in its speech, the sentiment, and whether the country had recognized

Palestine in that year. The predictor variable of our model is a binary variable, where 1 represents that the country voted in favor of Palestine, and 0 for any other choice.

Applying our model to speech data from 2016 to 2024 we get an f1 score of 53% in the imbalanced class, and overall accuracy of 83.5%. The performance was noticeably better for the majority class (Yes votes), where 1240 out of 1369 cases were predicted correctly (True Positives). On the other side, the model performed worse on the minority class (No/Other votes), correctly identifying 165 out of 313 cases (True Negatives) (Fig. 6).

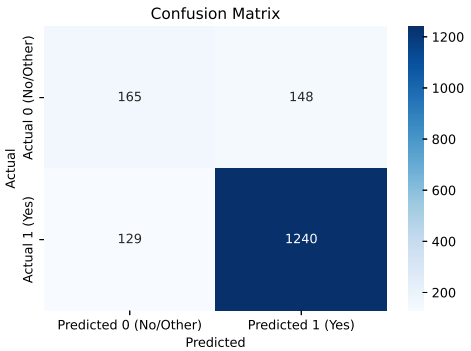


Fig. 5: Confusion Matrix on speeches in the period of 2016 - 2024

Due to the imbalance in the dataset with "Yes" being the majority, the model is reliable predicting votes that support Palestine but it struggles finding the opposition, which also may be driven by more complex geopolitical dynamics that are not well captured.

The feature importance analysis (Table. 2) provides further insight into the drivers of the model. The frequency of mentions about Palestine in a UN member's speech emerges as the most significant predictor, suggesting that giving importance to the topic by mentioning it, is a better indicator of voting behavior than the tone of the speech. Moreover, regions also contribute substantially, showing the importance of geopolitical groups in the UN (Europe, Asia, Oceania, and Africa), while sentiment (Positive, Neutral, Negative) and formal recognition play minimal role in the prediction of the model.

Although the predictive model was able to offer useful insights, several limitations must be acknowledged. One of the key limitations is the class imbalance that caused lower performance on the minority class (No/Other) which resulted in difficulty predicting the countries that do not vote in favor of Palestine. Moreover, broader geopolitical and economic variables could be explored to identify whether they could enrich the dataset and improve the ability of the model in accurately predict voting behavior. Finally, further work could be implemented by altering the sentiment model. Since VADER is trained on social media conversa-

Table 2: Feature Importance scores of Random Forest model

Feature	Score
palestine_keywords_count	29.5
Region_Name_Asia	5.7
Region_Name_Oceania	5.3
Region_Name_Africa	3.3
Sentiment_POSITIVE	1.9
Sentiment_NEGATIVE	1.4
recognized_palestine_True	1.3
recognized_palestine_False	1.2

tions, it cannot effectively capture the diplomatic sentiment as some statements can contain words with negative meaning while still expressing political support. The use of a BERT model (Devlin et al., 2019) fine-tuned on diplomatic lexicon, would be a good replacement of VADER for sentiment analysis, as found by Ma et al., 2019. This would result to more accurate scores in each UN General Debate speech, potentially making sentiment a more important explanatory factor of voting behavior.

4 Conclusion

The study explored whether patterns of Palestine mentions in UN General Debate speeches, together with region of the country, speech sentiment and Palestine state recognition, relate to UN members voting behavior on Palestine related resolutions. The exploratory analysis showed, that while Palestine is consistently mentioned throughout the years, descriptive factors were issue salience and geopolitical alignments rather than sentiment polarity and whether or not UN member states recognize a Palestinian state confirming partially our hypothesis.

On the predictive side, the question asked was whether speeches combined with the region of each UN member, sentiment, year and Palestine recognition could predict voting behavior. While the hypothesis was supported and the model predicted with an accuracy of 83.5%, the model performed worse in the minority class (No/Other) with an f1-score = 54%, mainly due to the imbalance in data as most UN member states vote in support of Palestine and the influence of other geopolitical factors that are not captured in the model.

Overall, the findings suggest that UN General Debate speeches do carry predictive importance, but primarily through issue discussion and geographic region rather than sentiment. These results highlight the complexity between alliances and diplomatic rhetoric, while also contributing to broader discussions on Sustainable Development Goal 16 on building peaceful and inclusive institutions. This could inform diplomatic strategy and lead to discussions on peace and human rights.

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Appendix A: Top TF-IDF word sentiment scores assigned to VADER using Senticnet

diplomatic 3.688 right -1.86 rights -1.86 must -0.932 diplomacy 3.304 exploitation -3.508 economic 3.588 efforts -0.68 development 3.584 including 0.692 resources 3.34 community 2.088 legitimate 3.524 nuclear -1.084 occupied -3.288 independent 3.392 settlement 1.316 continue 0.86 human 3.176 resolutions 3.444 lasting 3.308 selfdetermination 2.636 end -0.828 resolution 3.444 negotiations 3.308 establishment 2.38 work 3.468 full 3.316 without -1.196 force 3.304 recognized 1.508 cooperation 0.852 experience 3.832 recognition 0.736 still 2.54 inalienable 1.316 since -3.996 towards -2.028 explosive -3.348 living 3.572 borders -1.316 forces 3.304 deployed 3.496 homeland 0.832 cannot 3.396 part 3.876 possible 1.308 developing 3.584 believe 3.564 diplomat 3.688 refugees 3.608 principles 0.628 global -3.384 regard 3.788 establish 2.38 capital 3.356 deplore -3.54 concerned -0.896 issue -1.532 independence 3.392 explore 3.944 action 1.316 delegation 3.552 continues 0.86 population 0.676 democratic 3.44 representative 3.468 achieve 3.328 measures 3.284 high 0.756 powers 2.636

Appendix B: Voting around Palestine visual

Fig. 6: Percentage of Voting for Palestine per country Middle East (1969-2024)